

App Engine uses Google BigTable as the distributed storage system for persisting application data. If you want more information on BigTable and similar NoSQL databases, you can read my article on the subject in the February issue of *LINUX For You*.

The Eclipse environment

Google offers an Eclipse plug-in that provides both Google App Engine and GWT development capabilities. It also makes deploying and developing applications on App Engine easier. It can easily be installed from <http://dl.google.com/eclipse/plugin/3.6> via the Eclipse Update Manager.

To start a new App Engine application, click the new Web application project button, which will take you to a screen similar to what is shown in Figure 1. Fill in details like the project name, package and the instance of App Engine SDK to use. The Eclipse plug-in allows you to run applications for the Google App Engine locally, in an environment that simulates the App Engine environment, and is accessible at <http://localhost:8888/>. It has an admin console (like that in Figure 2), which allows you to view your *Local Datastore*, *Task Queues*, *Inbound Mail*, *XMPP*, *Backends*, etc, at http://localhost:8888/_ah/admin/. The Eclipse environment creates a default directory structure resembling the one shown in Figure 3.

Deploying applications

To deploy your application to Google servers, you need an App Engine account or a Gmail account. Go to <http://appengine.google.com> and log in with your Google account information. After that, you will have to confirm your mobile number by an SMS sent to your number, and then you can create your application by providing a unique

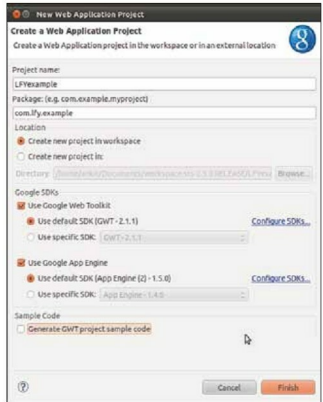


Figure 1: The new Web application dialog box

application identifier and application title. Now, enter these details in the Eclipse plug-in to enable direct deployment from there. To set up deployment for your application to App Engine, you need to mention your application ID in the file '*war/WEB-INF/appengine-web.xml*' mentioned under the tag application, which you had created during the App Engine registration. To illustrate this, let's take an "LFYExample" application:

```
<?xml version="1.0" encoding="utf-8"?> <appengine-
```



Figure 2: The local Admin console